Walchand College of Engineering, Sangli

Dept. of Computer Science and Engineering, TY (A)

MINI PROJECT-I (7CS345) — LA-2 EVALUATION, 2026-27

PROMPTOPT - TOKEN-EFFICIENT PROMPT OPTIMIZATION FOR LLMS

Team Members

Piush U. Gogi - 255200005

Nirzara A. Manade - 255200006

Siddhi S. Bolaikar - 255200008

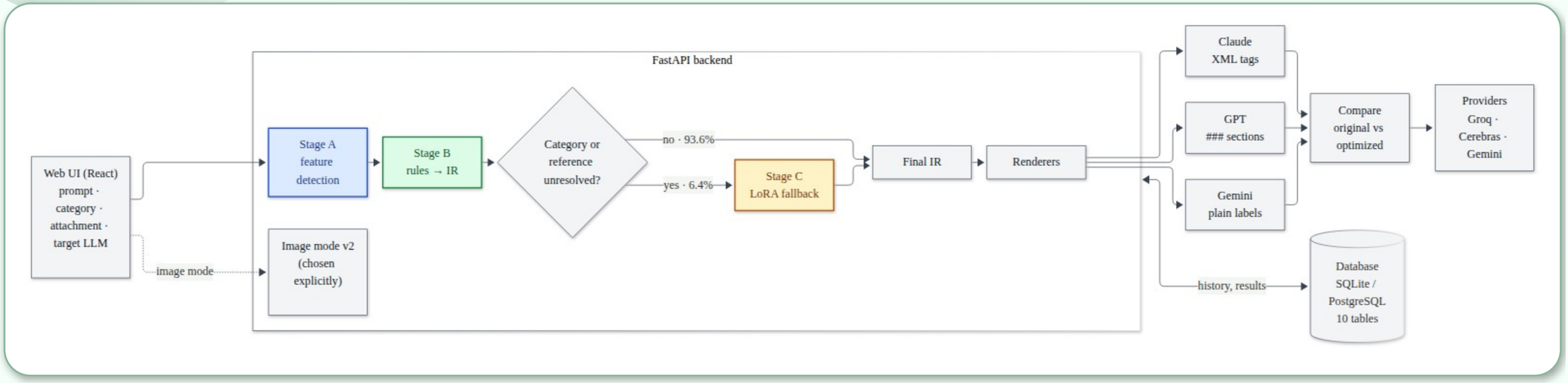
Under the Guidance of -

Prof. A.S. Pawar

AGENDA

1. Core Modules
2. Transaction Flow & Database
3. Unit Testing
4. Integration & System Testing
5. Test Cases & Correctness
6. Results & Working Prototype
7. Git & GitHub
8. Challenges & Future Plans

SYSTEM ARCHITECTURE



Stage A

spaCy + regex + sentence-embedding classifier: category, missing format and constraints, filler, ambiguous references

Stage B

15 deterministic rules B01-B15, every change logged; category rules only at confidence ≥ 0.6

Stage C

Qwen2.5-0.5B + LoRA, called only for unresolved fields; validated, else Stage B is kept

Output

IR rendered per target LLM with token counts; Compare, History, 30-day retention

STAGE A DETECTS, STAGE B FIXES

STAGE A · FEATURE DETECTION

- Task category + confidence (embedding classifier, keyword fallback)
- A02 missing output format
- A03 missing length / tone / audience / language
- A04 filler and repetition
- A05 ambiguous reference ("summarize this")

74.7%

category accuracy on 482 test prompts (coding F1 0.98, classification 0.94)

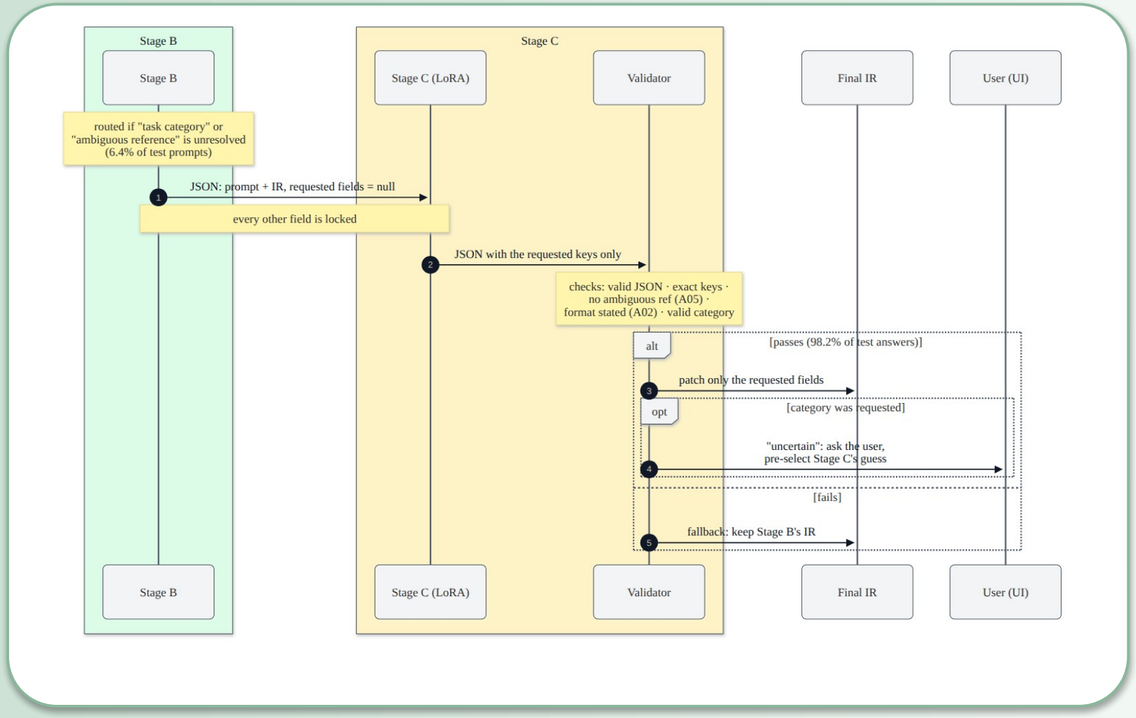
STAGE B · 15 RULES, EACH A UNIT-TESTED FUNCTION

B01 Remove filler	**B02** Remove duplicates	**B03** Output format
B04 Length limit	**B05** Programming language	**B06** Label set
B07 Structure / data block	**B08** Group fallback	**B09-B15** Attachment modifiers

3.1% → 95.0%

of test prompts state an output format after Stage B; 6.4% (31 of 482) are routed to Stage C

STAGE C: LORA ONLY WHERE RULES STOP



B → C contract: Stage C gets only the unresolved fields, must return JSON with exactly those keys, is checked with Stage A's own detectors, and falls back to Stage B if invalid.

5% → 98%

answers pass validation: base model vs LoRA

22.6 → 90.3%

format stated on routed prompts

0.74 s

median per call on RTX 3050 (CPU 4.55 s)

6.4%

of test prompts routed to Stage C

Forced on every prompt, Stage C lowers task intent (0.886 → 0.769), so it stays a fallback. Its category is only a suggestion: the UI asks the user and pre-selects it.

IR, RENDERERS, ATTACHMENTS, IMAGE MODE

Claude

```
<task>Write code to get
all permutations of
a string.</task>
<constraints>
- Use Python.
</constraints>
<output_format>
Return only the code
</output_format>
```

GPT

```
### Task
Write code to get
all permutations of
a string.
### Constraints
- Use Python.
### Output format
Return only the code
```

Gemini

```
Task: Write code to
get all permutations
of a string.
Constraints:
Use Python.
Output format:
Return only the code
```

Same content, layout per model · every rendering is parsed back in the tests · GPT tokens exact (tiktoken)

Attachments B09-B15: image, PDF, PPTX, DOCX, spreadsheet, code, other; fire only with an attachment · **30/30** hand-made prompts correct

IMAGE MODE (SEPARATE, USER-SELECTED)

Held-out CLIP vs the user's own prompt (30 prompts)

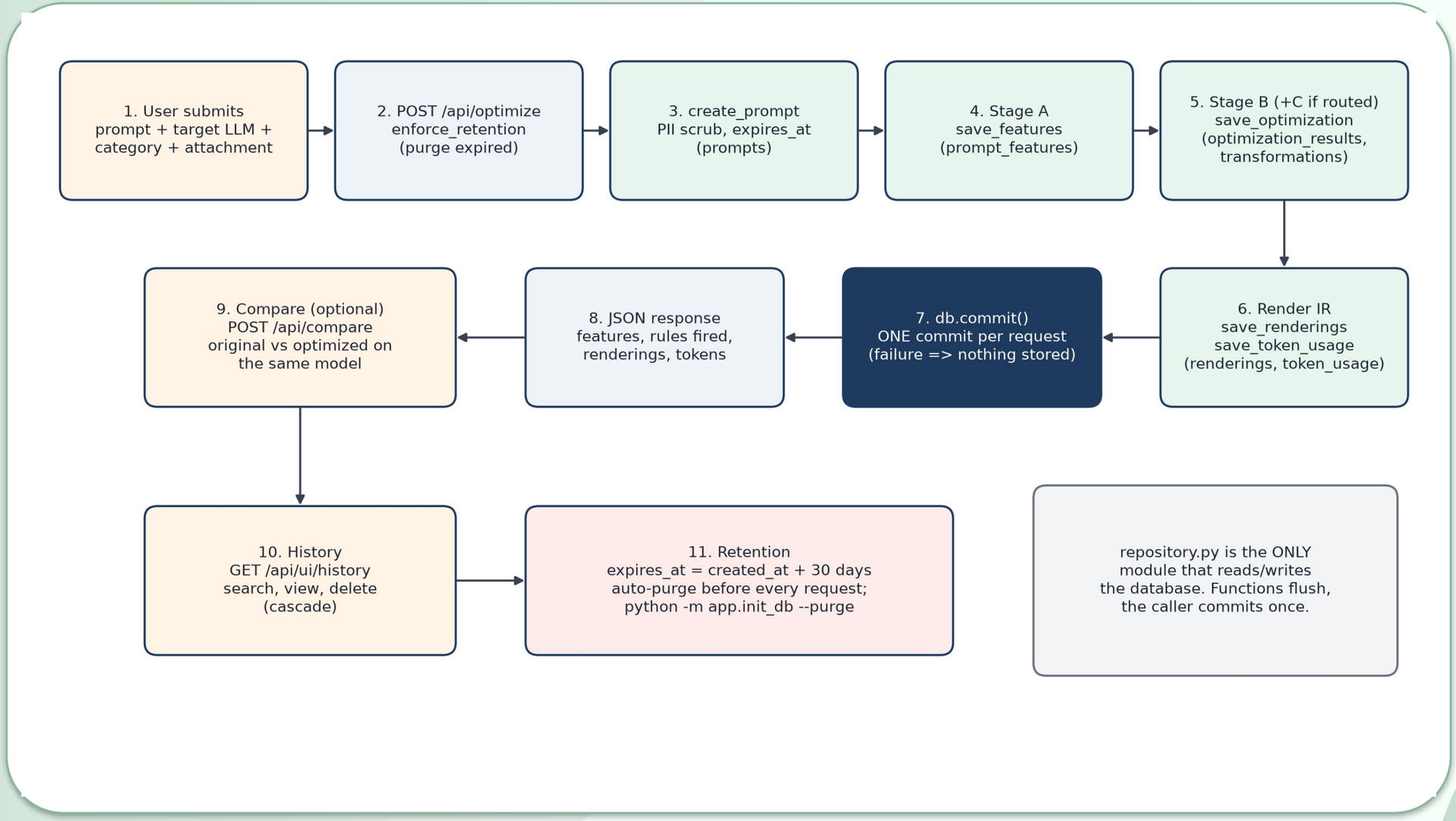
Original
33.08

v1: defaults
30.78 · worse, p = 0.043

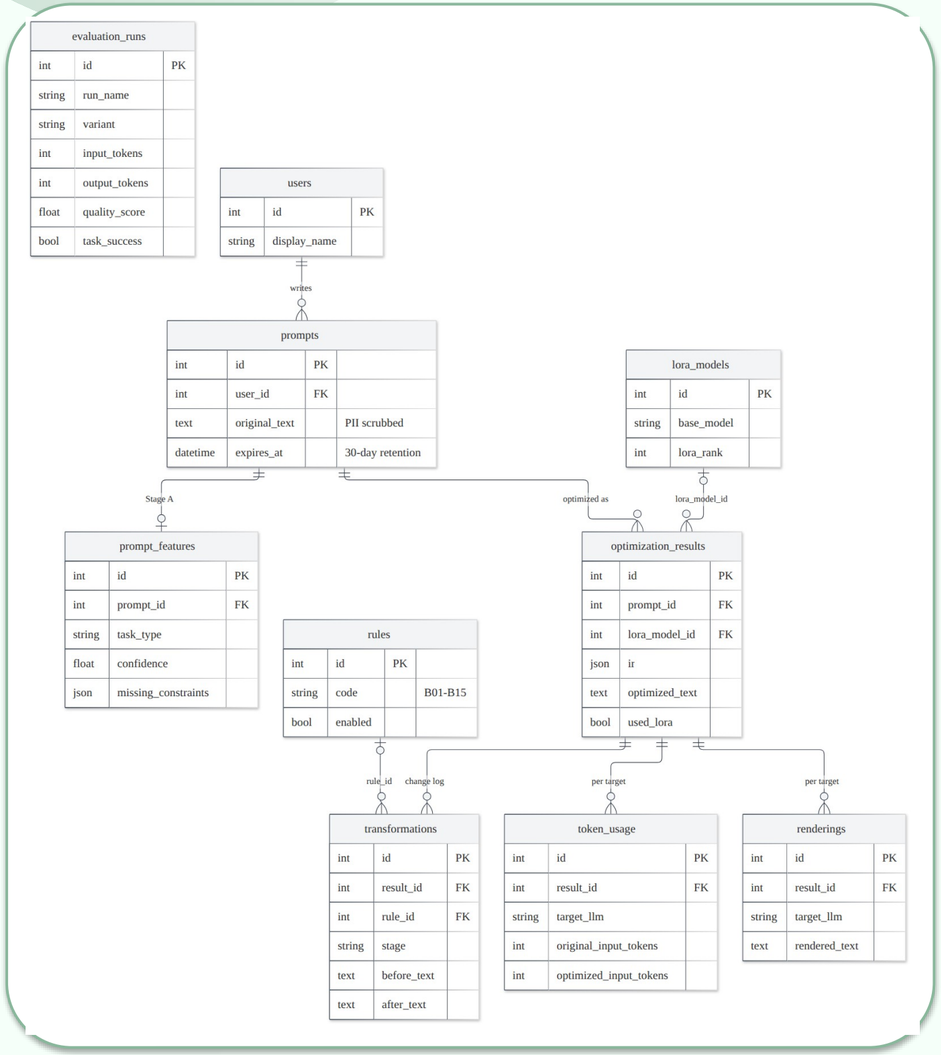
v2: suggestions
32.77 · no measurable harm

Style kept in 12/12 styled prompts with v2

TRANSACTION FLOW



DATABASE LAYER: 10 TABLES

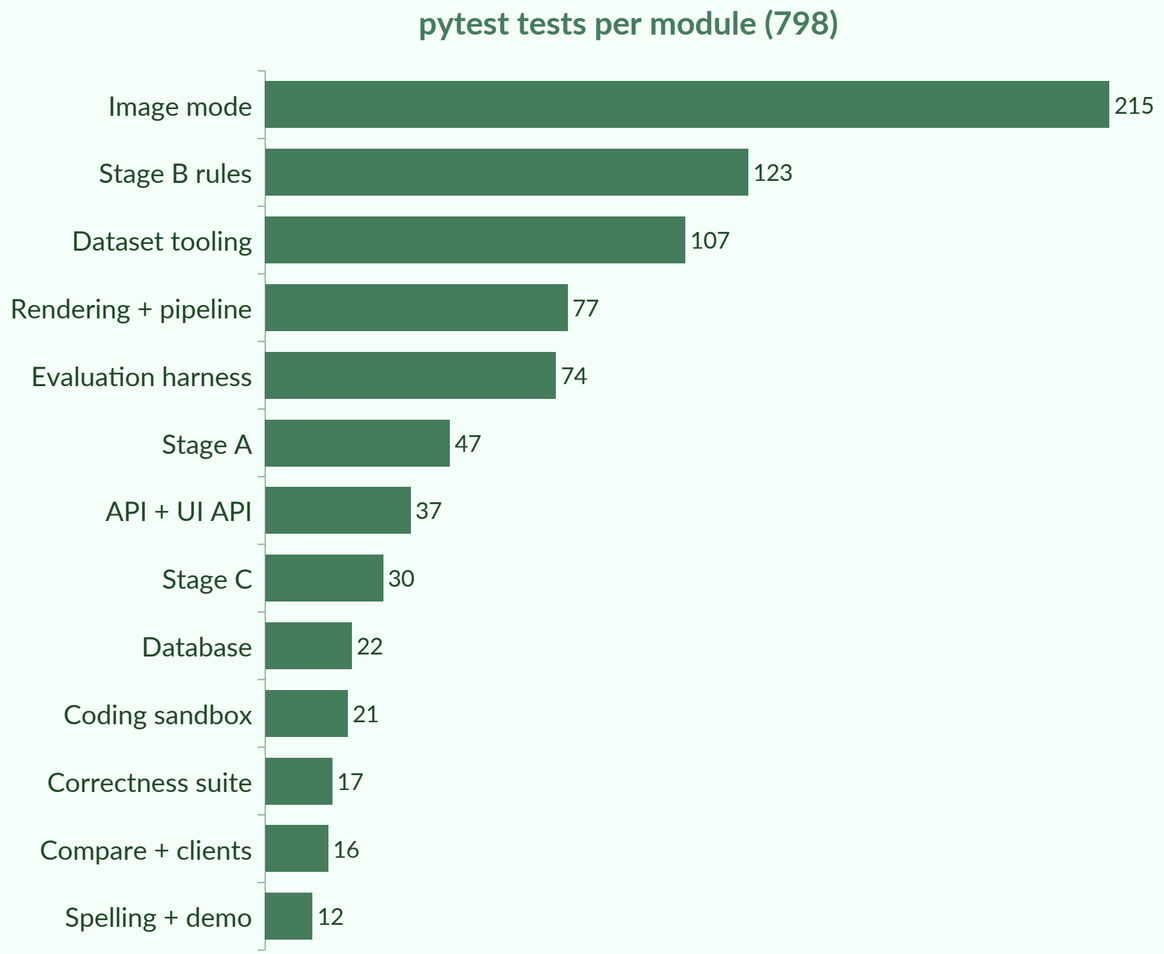


WRITTEN IN ONE OPTIMIZE TRANSACTION (VIA REPOSITORY.PY)

Table	Repository call	Stores
prompts	create_prompt	PII-scrubbed text, expires_at
prompt_features	save_features	category, confidence, gaps
optimization_results	save_optimization	IR, optimized text, used_lora
transformations	save_optimization	one row per change, rule_id
renderings	save_renderings	prompt per target LLM
token_usage	save_token_usage	tokens per target

- Cascade on delete: purge removes everything derived from a prompt
- PII (e-mail, phone) scrubbed before storing
- 30-day retention enforced before every request · SQLite / PostgreSQL

UNIT TESTING: 798 TESTS, ALL PASSING



RUN

```
cd backend
source .venv/bin/activate
python -m pytest -q
→ 798 passed in 24.6 s
```

REAL EXAMPLES

- B05 defaults to Python
- PII scrubbed before storage
- retention purge cascades
- blank prompt → HTTP 422, not 500
- invalid Stage C answer → Stage B kept
- sandbox has no network

INTEGRATION AND SYSTEM TESTING

FREEZE CHECK

BYTE-IDENTICAL

482 test prompts, Stage A+B vs tag frozen-for-test (sha256 35f7d4dc…)

UI END-TO-END

13 / 13

Playwright + Chrome: optimize, compare, suite, history, image mode, no console errors

API END-TO-END

pytest

FastAPI TestClient: Stage C on/off, 422 errors, retention, PII-scrubbed Compare

FULL PIPELINE

482 + 44

test split run once (2 Oct) · benchmark with a real LLM + blind judge

CODING SANDBOX

20.7 → 34.5%

pass@1 strict, 29 Python items; bubblewrap: no network, read-only

CORRECTNESS SUITE

37 → 42 / 50

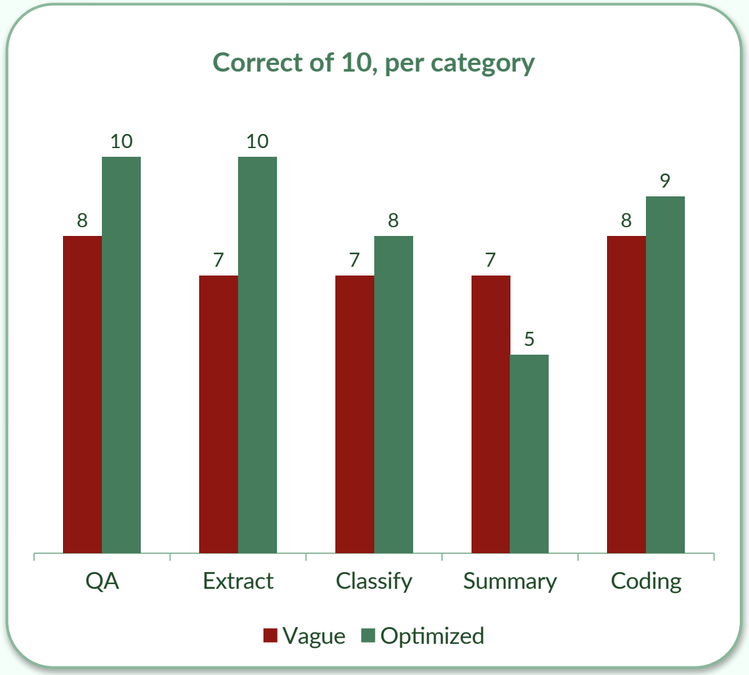
hand-written cases with verifiable answers, gold human-reviewed

TEST CASES AND CORRECTNESS

ID	Case	Input	Result	Status
TC04	B01 removes filler	"hey can you please summarize the text for me?"	"Summarize the text."	PASS
TC16	PII scrubbed	mail rahul.k@example.com … 98765 43210	1 [EMAIL], 2 [PHONE]	PASS
TC26	Freeze check	482 test prompts	byte-identical	PASS
TC31	cqa-01 hostel fee	"if i pay on 18 july how much total??"	80300 (vague said 85300)	PASS
TC34	cls-09 attendance	"mark each student safe warning or detained"	got a Python script	FAIL
TC37	cod-10 recipe scaler	"write the scale fn for recipe app"	4/8 asserts	FAIL

Fixed 9 · broke 4 · McNemar p = 0.267 (not significant at n = 50)

Failures: Stage A misread a table as code (cls-09); two summaries 2-3 words over the limit; one Stage C coding case passed 4 of 8 asserts.



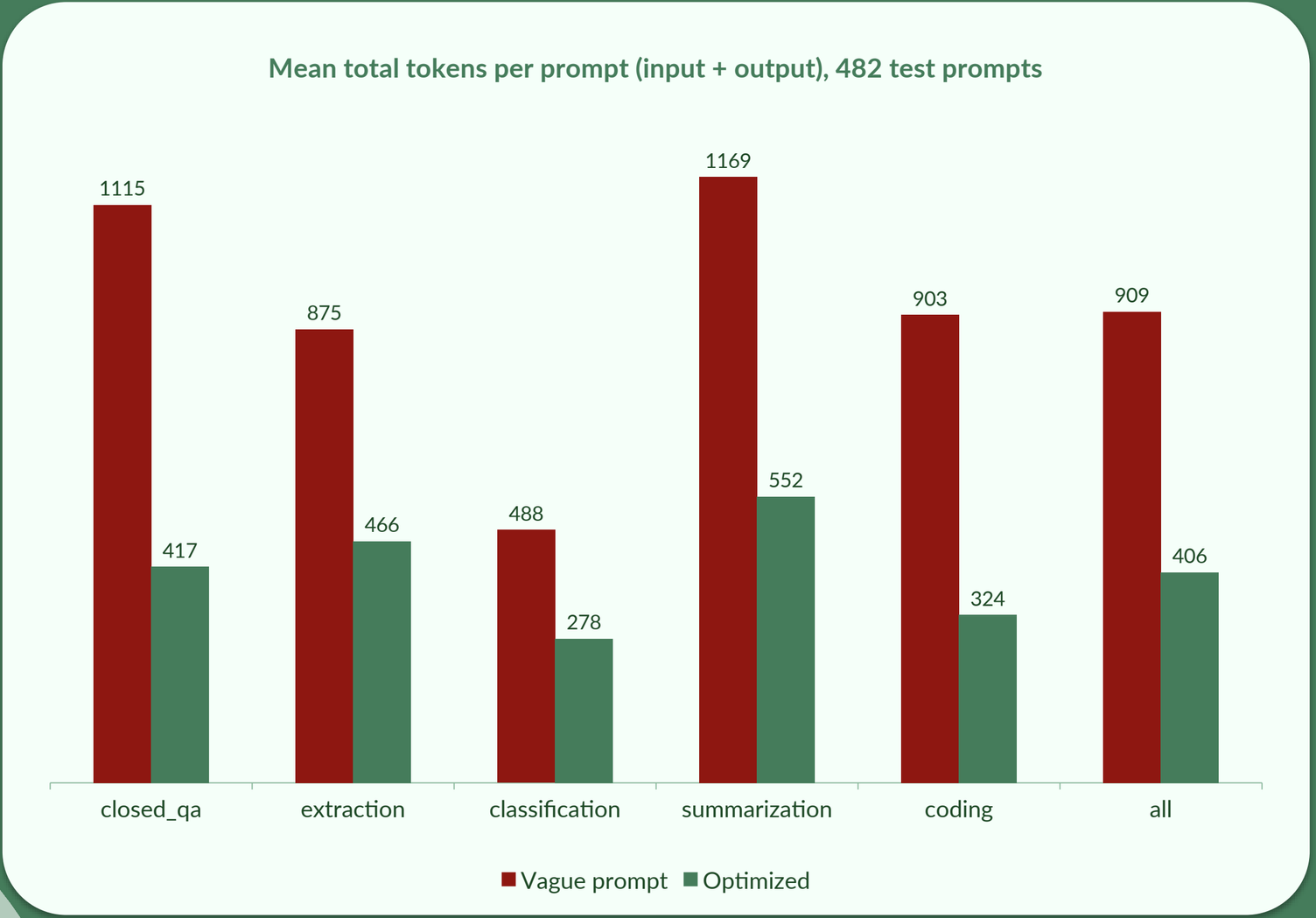
42/50

optimized (vague 37)

-29.6%

total tokens

RESULTS: -39.2% TOTAL TOKENS



682 → 165

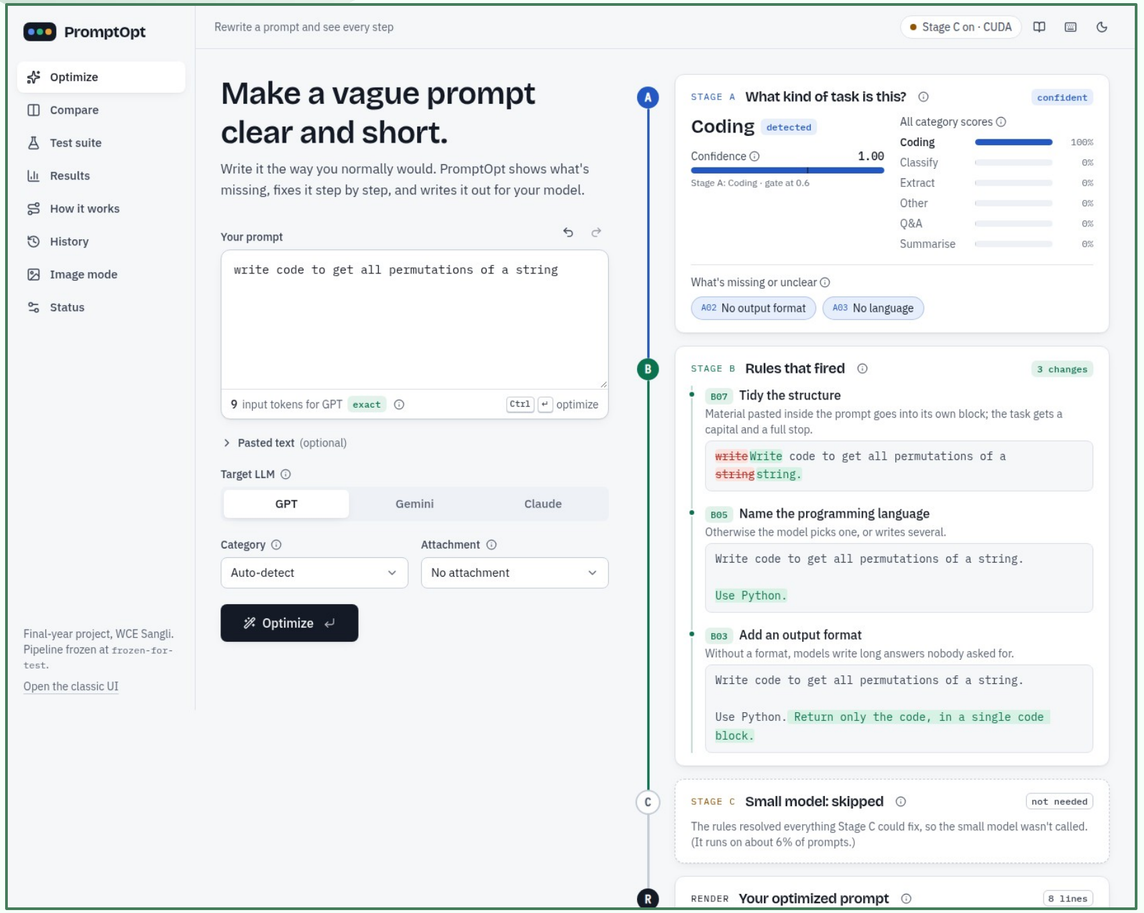
output tokens per prompt (input 226 → 241)

8.2 → 9.0

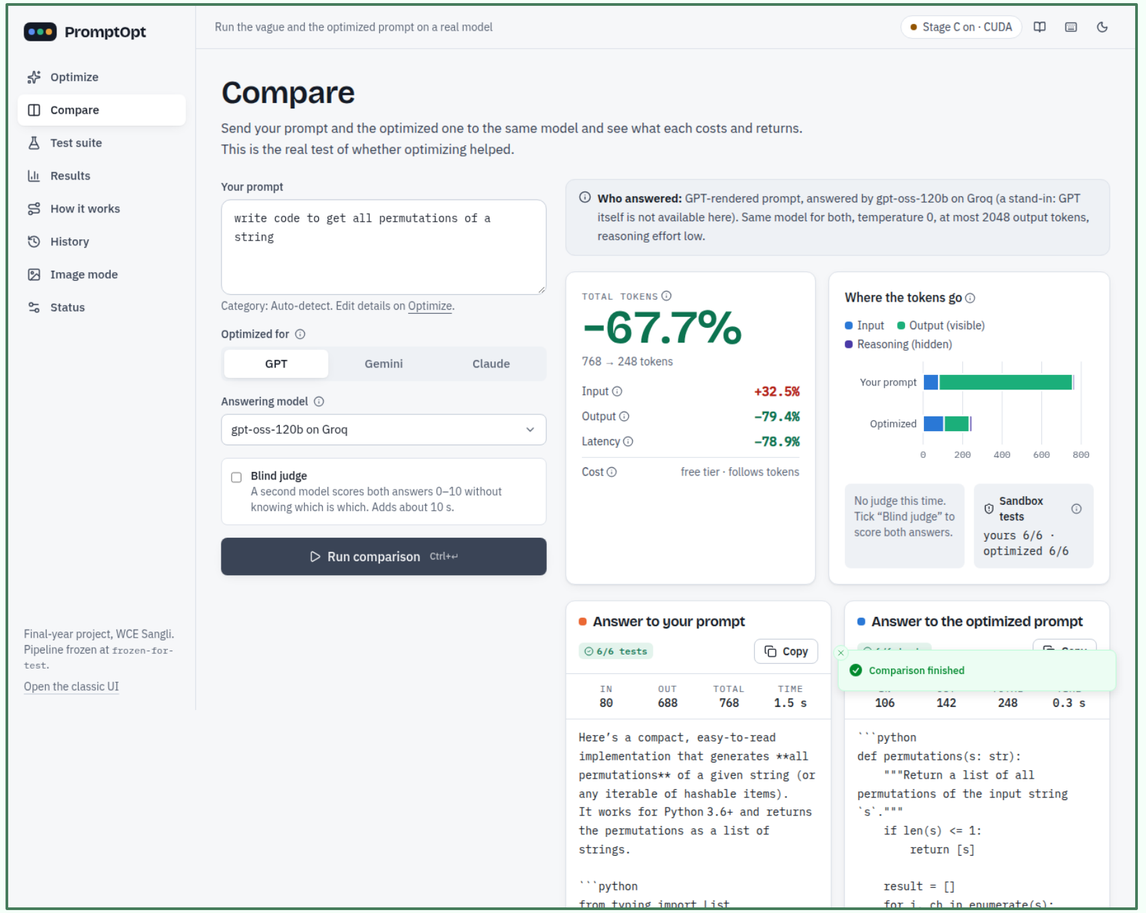
benchmark quality; task success 67% → 85%

Limitation 87 of 482 prompts cost more: their answer was already short (mostly extraction). Next: lean mode.

WORKING PROTOTYPE

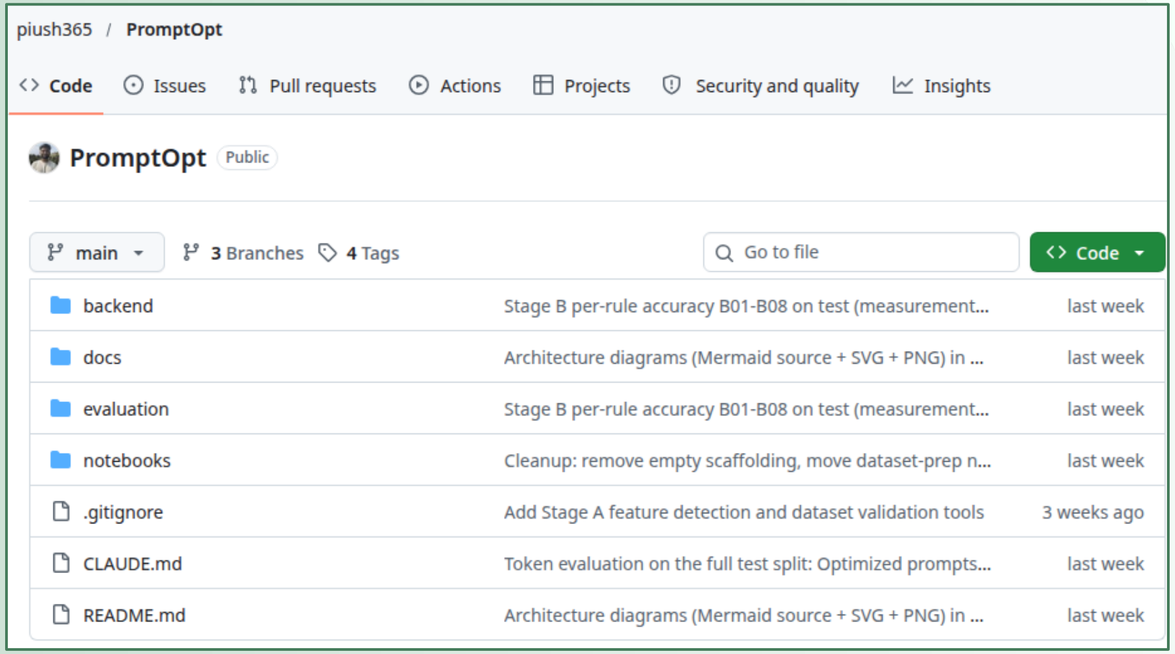


Optimize: every stage shown, rule by rule, with before/after text and the rendered prompt



Compare: 768 → 248 tokens (-67.7%) on gpt-oss-120b, both answers pass 6/6 tests

GIT AND GITHUB



github.com/piush365/PromptOpt

Tags are our evaluation checkpoints: after frozen-for-test no Stage A/B rule, threshold or judge changed, and the freeze check proves it.

92
commits

3
branches

4
tags · 1 release

- frozen-for-test (27 Sep): code for the final numbers
- final-for-test (2 Oct): test split run once
- v1.0 (8 Oct): release with the Stage C adapter
- Commits per month: Aug 7 · Sep 38 · Oct 47
- Tracking: roadmap and phase tables in CLAUDE.md, tags, release (no GitHub Issues/Projects used)

CHALLENGES AND HOW WE SOLVED THEM

DATASET GENERATION BUGS

LLM targets dropped data → deterministic repair: 349 prompts fixed, every change logged

KAPPA PARADOX

90% agreement but kappa 0.065 → report Gwet's AC1 (0.92-0.96) next to it

API QUOTA LIMITS

Groq 200k tokens/day → rolling 24-h usage ledger, 70% budget rule, resumable runs

STAGE C HALLUCINATION

base model 5% valid → LoRA 98.2%, strict JSON contract, fallback to Stage B

IMAGE MODE V1 FAILURE

defaults turned a watercolor into a photo → v2: suggestions only

LABEL / REGEX EDGE CASES

"thank you so much" inside a button → a unit test per edge case, B16 in the app

4 GB GPU TRAINING

micro-batch 1 × 16, bf16, gradient checkpointing: 51 min; models loaded once

REPORT IT, DON'T HIDE IT

After the freeze, bugs were reported, not fixed; failures are shown in the results.

PLANS FOR THE UPCOMING PHASES

1. **Lean mode**
   add only the format line when the answer is short
2. **Smarter routing**
   send label-set and typo prompts to Stage C
3. **Spelling normalisation**
   user-confirmed fixes before optimizing
4. **Blind user tests**
   prompts written by people outside the team
5. **Real GPT / Claude / Gemini**
   Compare with paid API keys
6. **Larger correctness suite**
   more cases and providers for significance
7. **Deployment**
   PostgreSQL, HTTPS and login, beyond the LAN

LIVE DEMO

1. **Optimize** "write code to get all permutations of a string"
2. **Stage C** "hey can you just summarize this article for me"
3. **Compare** gpt-oss-120b: tokens, latency, sandbox tests
4. **Image mode** "oil painting of a sailboat in a storm"
5. **History · Test suite · Results** past prompts, 50-case suite, headline numbers

THANK YOU